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```
In [302... import pandas as pd
import numpy as np
```

Q1. Create a DataFrame containing student information like Name, Age, Department, %Marks

```
In [303... data = {
    'Name': ['aa', 'bb', 'cc', 'dd'],
    'age' : [18, 21, 22, 22],
    'department': ['cse', 'ece', 'eee', 'cse'],
    'marks' : [85, 75, 65, 95]
}
df = pd.DataFrame(data)
df
```

```
Out[303...      Name  age  department  marks
0      aa   18         cse     85
1      bb   21         ece     75
2      cc   22         eee     65
3      dd   22         cse     95
```

a. display the column names of the df

```
In [304... cols = df.columns
cols
```

```
Out[304... Index(['Name', 'age', 'department', 'marks'], dtype='object')
```

b. Show min/max and average marks

```
In [305... maximum = (df['marks'].max())
minimum = (df['marks'].min())
avg = (df['marks'].mean())
print(f'avg is {avg}, max is {maximum}, min is {minimum}')
```

avg is 80.0, max is 95, min is 65

c. Get unique values and student count for each Department

```
In [306... unique_dept = df['department'].unique()
unique_dept
```

```
Out[306... array(['cse', 'ece', 'eee'], dtype=object)
```

```
In [307... d = df.groupby('department')['Name'].count()
```

d

```
Out[307...] department
cse      2
ece      1
eee      1
Name: Name, dtype: int64
```

Q2. Read csv file solar.csv

```
In [308...] df = pd.read_csv('solar.csv')
df
```

```
Out[308...]

```

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
0	California	289	4395	15.3	10826	West
1	Arizona	48	1078	22.5	2550	West
2	Nevada	11	238	21.6	557	West
3	Chicago	33	261	7.9	590	North
4	Connecticut	20	118	5.9	235	East
5	Texas	12	187	15.6	354	South
6	New York	13	53	4.1	84	East
7	Florida	15	535	12.3	453	South

a. Display all the plants with capacity > 500 MW

```
In [309...] q1 = df[df['Installed Capacity (MW)'] > 500]
q1
```

```
Out[309...]

```

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
0	California	289	4395	15.3	10826	West
1	Arizona	48	1078	22.5	2550	West
7	Florida	15	535	12.3	453	South

b. Display plant details for New York plant

```
In [310...] q2 = df[df['State'] == 'New York']
q2
```

Out[310...

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
6	New York	13	53	4.1	84	East

c. For all the plants display Average MW Per Plant and Generation (GWh)

In [311...

```
#to give avg mw per plant, and generation(GWh)
q3 = df[['Average MW Per Plant', 'Generation (GWh)']]
q3
```

Out[311...

	Average MW Per Plant	Generation (GWh)
0	15.3	10826
1	22.5	2550
2	21.6	557
3	7.9	590
4	5.9	235
5	15.6	354
6	4.1	84
7	12.3	453

d. Sort the plants by Generation (GWh) (ascending) and Installed Capacity (MW)(descending)

In [312...

```
df.sort_values(by=['Generation (GWh)', 'Installed Capacity (MW)'], ascending=[True, False])
```

Out[312...

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
6	New York	13	53	4.1	84	East
4	Connecticut	20	118	5.9	235	East
5	Texas	12	187	15.6	354	South
7	Florida	15	535	12.3	453	South
2	Nevada	11	238	21.6	557	West
3	Chicago	33	261	7.9	590	North
1	Arizona	48	1078	22.5	2550	West
0	California	289	4395	15.3	10826	West

e. List top 5 plants for their Generation

```
In [313... df.sort_values(by = 'Generation (GWh)', ascending=False).head(5)
```

Out[313...

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
0	California	289	4395	15.3	10826	West
1	Arizona	48	1078	22.5	2550	West
3	Chicago	33	261	7.9	590	North
2	Nevada	11	238	21.6	557	West
7	Florida	15	535	12.3	453	South

f. Display all plant details for states – California, Nevada, Arizona and Texas

```
In [314... states = ['California', 'Nevada', 'Arizona', 'Texas']
ifiltered = df.loc[df['State'].isin(states)]
ifiltered
# df['State'] ==
```

Out[314...

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
0	California	289	4395	15.3	10826	West
1	Arizona	48	1078	22.5	2550	West
2	Nevada	11	238	21.6	557	West
5	Texas	12	187	15.6	354	South

g. Group the plants by region, find – min/max capacity plant for each group

```
In [315... grouped = df.groupby('Region').agg(
    min_capacity = ('Installed Capacity (MW)', 'min'),
    max_capacity = ('Installed Capacity (MW)', 'max')
)
print(grouped)
```

	min_capacity	max_capacity
Region		
East	53	118
North	261	261
South	187	535
West	238	4395

h. Add following information to the existing details for the plants

```
In [316... df.loc[len(df)] = ['North Carolina', 148, 669, 4.5, 1162, 'North' ]
df
```

Out[316...

	State	Number of Solar Plants	Installed Capacity (MW)	Average MW Per Plant	Generation (GWh)	Region
0	California	289	4395	15.3	10826	West
1	Arizona	48	1078	22.5	2550	West
2	Nevada	11	238	21.6	557	West
3	Chicago	33	261	7.9	590	North
4	Connecticut	20	118	5.9	235	East
5	Texas	12	187	15.6	354	South
6	New York	13	53	4.1	84	East
7	Florida	15	535	12.3	453	South
8	North Carolina	148	669	4.5	1162	North

Q.3Read csv gapminder2007.csv

In [317...

```
gap = pd.read_csv('gapminder2007.csv')
gap
```

Out[317...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

a. Display data for all Asian countries

In [318...

```
gap[gap['continent'] == 'Asia']
```

Out[318...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
7	Bahrain	708573	Asia	75.635	29796.048340
8	Bangladesh	150448339	Asia	64.062	1391.253792
18	Cambodia	14131858	Asia	59.723	1713.778686
24	China	1318683096	Asia	72.961	4959.114854
55	Hong Kong, China	6980412	Asia	82.208	39724.978670
58	India	1110396331	Asia	64.698	2452.210407
59	Indonesia	223547000	Asia	70.650	3540.651564
60	Iran	69453570	Asia	70.964	11605.714490
61	Iraq	27499638	Asia	59.545	4471.061906
63	Israel	6426679	Asia	80.745	25523.277100
66	Japan	127467972	Asia	82.603	31656.068060
67	Jordan	6053193	Asia	72.535	4519.461171
69	Korea, Dem. Rep.	23301725	Asia	67.297	1593.065480
70	Korea, Rep.	49044790	Asia	78.623	23348.139730
71	Kuwait	2505559	Asia	77.588	47306.989780
72	Lebanon	3921278	Asia	71.993	10461.058680
78	Malaysia	24821286	Asia	74.241	12451.655800
83	Mongolia	2874127	Asia	66.803	3095.772271
87	Myanmar	47761980	Asia	62.069	944.000000
89	Nepal	28901790	Asia	63.785	1091.359778
96	Oman	3204897	Asia	75.640	22316.192870
97	Pakistan	169270617	Asia	65.483	2605.947580
101	Philippines	91077287	Asia	71.688	3190.481016
109	Saudi Arabia	27601038	Asia	72.777	21654.831940
113	Singapore	4553009	Asia	79.972	47143.179640
119	Sri Lanka	20378239	Asia	72.396	3970.095407
124	Syria	19314747	Asia	74.143	4184.548089
125	Taiwan	23174294	Asia	78.400	28718.276840
127	Thailand	65068149	Asia	70.616	7458.396327

	country	pop	continent	lifeExp	gdpPercap
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906

b. Display top and bottom 15 rows

```
In [319... top_bottom_15 = pd.concat([gap.head(15), gap.tail(15)])  
top_bottom_15
```

Out[319...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
5	Australia	20434176	Oceania	81.235	34435.367440
6	Austria	8199783	Europe	79.829	36126.492700
7	Bahrain	708573	Asia	75.635	29796.048340
8	Bangladesh	150448339	Asia	64.062	1391.253792
9	Belgium	10392226	Europe	79.441	33692.605080
10	Benin	8078314	Africa	56.728	1441.284873
11	Bolivia	9119152	Americas	65.554	3822.137084
12	Bosnia and Herzegovina	4552198	Europe	74.852	7446.298803
13	Botswana	1639131	Africa	50.728	12569.851770
14	Brazil	190010647	Americas	72.390	9065.800825
127	Thailand	65068149	Asia	70.616	7458.396327
128	Togo	5701579	Africa	58.420	882.969944
129	Trinidad and Tobago	1056608	Americas	69.819	18008.509240
130	Tunisia	10276158	Africa	73.923	7092.923025
131	Turkey	71158647	Europe	71.777	8458.276384
132	Uganda	29170398	Africa	51.542	1056.380121
133	United Kingdom	60776238	Europe	79.425	33203.261280
134	United States	301139947	Americas	78.242	42951.653090
135	Uruguay	3447496	Americas	76.384	10611.462990
136	Venezuela	26084662	Americas	73.747	11415.805690
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

c. Show all the rows where life expectancy is more than 50 years and less than 80 years

```
In [320... filtered = gap[(gap['lifeExp'] > 50) & (gap['lifeExp'] < 80)]
filtered
```

```
Out[320...
country    pop  continent  lifeExp  gdpPercap
1  Albania  3600523    Europe   76.423   5937.029526
2  Algeria  33333216     Africa   72.301   6223.367465
4  Argentina 40301927  Americas   75.320  12779.379640
6  Austria   8199783    Europe   79.829  36126.492700
7  Bahrain   708573      Asia    75.635  29796.048340
...      ...      ...      ...      ...
135 Uruguay   3447496  Americas   76.384  10611.462990
136 Venezuela 26084662  Americas   73.747  11415.805690
137 Vietnam   85262356    Asia    74.249   2441.576404
138 West Bank and Gaza 4018332    Asia    73.422   3025.349798
139 Yemen, Rep. 22211743    Asia    62.698   2280.769906
```

110 rows × 5 columns

d. Show data for India, America and France

```
In [321... countries = ['India', 'America', 'France']
gap[gap['country'].isin(countries)]
```

```
Out[321...
country    pop  continent  lifeExp  gdpPercap
44  France  61083916    Europe   80.657  30470.016700
58   India 1110396331    Asia    64.698   2452.210407
```

e. Show data for all countries where name starts with 'A'

```
In [322... gap[gap['country'].str.match('A.*')] ]
```

Out[322...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
5	Australia	20434176	Oceania	81.235	34435.367440
6	Austria	8199783	Europe	79.829	36126.492700

f. Show min/max and mean GDP and Population for each continent

In [323...

```
min_max = gap.groupby('continent').agg(
    min_gdp = ('gdpPercap', 'min'),
    max_gdp = ('gdpPercap', 'max'),
    mean_gdp = ('gdpPercap', 'mean'),
)
```

min_max

Out[323...

	min_gdp	max_gdp	mean_gdp
continent			
Africa	277.551859	13206.48452	3089.032605
Americas	1201.637154	42951.65309	11003.031625
Asia	944.000000	47306.98978	12473.026870
Europe	5937.029526	49357.19017	25054.481636
Oceania	25185.009110	34435.36744	29810.188275

g. Show population and life expectancy for all countries

In [324...

gap

Out[324...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

h. Sort the data by country names (ascending) and GDP (descending)

In [325...

```
gap.sort_values(by=['country', 'gdpPercap'], ascending=[True, False])
```

Out[325...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

i. Find top 20 most populous countries

In [326... `gap.sort_values(by='pop', ascending=False).head(20)`

Out[326...

	country	pop	continent	lifeExp	gdpPercap
24	China	1318683096	Asia	72.961	4959.114854
58	India	1110396331	Asia	64.698	2452.210407
134	United States	301139947	Americas	78.242	42951.653090
59	Indonesia	223547000	Asia	70.650	3540.651564
14	Brazil	190010647	Americas	72.390	9065.800825
97	Pakistan	169270617	Asia	65.483	2605.947580
8	Bangladesh	150448339	Asia	64.062	1391.253792
94	Nigeria	135031164	Africa	46.859	2013.977305
66	Japan	127467972	Asia	82.603	31656.068060
82	Mexico	108700891	Americas	76.195	11977.574960
101	Philippines	91077287	Asia	71.688	3190.481016
137	Vietnam	85262356	Asia	74.249	2441.576404
47	Germany	82400996	Europe	79.406	32170.374420
38	Egypt	80264543	Africa	71.338	5581.180998
42	Ethiopia	76511887	Africa	52.947	690.805576
131	Turkey	71158647	Europe	71.777	8458.276384
60	Iran	69453570	Asia	70.964	11605.714490
127	Thailand	65068149	Asia	70.616	7458.396327
27	Congo, Dem. Rep.	64606759	Africa	46.462	277.551859
44	France	61083916	Europe	80.657	30470.016700

j. Delete all rows for a specific country

In [327... `condition = gap['country'] == 'England'`
`dropped_indices = gap[condition].index`
`gap.drop(dropped_indices, inplace=True)`
`gap`

Out[327...

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

k. Change all column names to title case

In [328...

```
gap.columns = list(map(lambda x : x.title(), gap.columns))
gap
```

Out[328...

	Country	Pop	Continent	Lifeexp	Gdppercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

l. Change column names: pop to Population, gdpPercap to GDP_Percap

```
In [329... gap.rename(columns = { 'Pop' : 'Population', 'Gdppercap': 'GDP_Percap' }, inplace=True)
gap
```

```
Out[329...
      Country  Population  Continent  Lifeexp  GDP_Percap
0  Afghanistan    31889923        Asia    43.828    974.580338
1    Albania      3600523        Europe    76.423    5937.029526
2    Algeria     33333216        Africa    72.301    6223.367465
3    Angola      12420476        Africa    42.731    4797.231267
4  Argentina     40301927    Americas    75.320   12779.379640
...         ...         ...         ...         ...         ...
137  Vietnam      85262356        Asia    74.249   2441.576404
138  West Bank and Gaza    4018332        Asia    73.422   3025.349798
139  Yemen, Rep.     22211743        Asia    62.698   2280.769906
140    Zambia      11746035        Africa    42.384   1271.211593
141    Zimbabwe     12311143        Africa    43.487    469.709298
```

142 rows × 5 columns

m. Increase Life Expectancy by 2 years for all African countries

```
In [330... gap.loc[gap['Continent'] == 'Africa', 'Lifeexp'] +=2
gap
```

Out[330...

	Country	Population	Continent	Lifeexp	GDP_Percap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	74.301	6223.367465
3	Angola	12420476	Africa	44.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	44.384	1271.211593
141	Zimbabwe	12311143	Africa	45.487	469.709298

142 rows × 5 columns

n. Find all the rows that contain NA values

here we have no null values hence we get 0 as the sum value and all values as false

In [331...

```
na_values = gap.isna().any(axis = 0).sum()
print(na_values)

gap.isna().any(axis = 0)
```

0

Out[331...

```
Country      False
Population   False
Continent    False
Lifeexp      False
GDP_Percap   False
dtype: bool
```

o. Fill NA values with 0 for numeric columns

In [332...

```
gap.fillna(0)
```

Out[332...

	Country	Population	Continent	Lifeexp	GDP_Percap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	74.301	6223.367465
3	Angola	12420476	Africa	44.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	44.384	1271.211593
141	Zimbabwe	12311143	Africa	45.487	469.709298

142 rows × 5 columns